CampusIQ - Day 1 Notes

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Topic: Laptop setup, Git, GitHub and Python basics

1. Key Words

- **Git** = a tool that saves our code and keeps its history.
- **GitHub** = the online home for our code (like Google Drive).
- **Repository (repo)** = the project folder.
- **Commit** = a save point.
- **Push** = upload code from laptop to GitHub.
- **Pull** = download the latest code from GitHub to laptop.
- **Clone** = copy the repo from GitHub to our laptop (done only once).
- **Python** = our main language. It is simple, almost like English.

2. Tools to Install

1. **Python 3.11** (python.org). **Tick "Add Python to PATH" while installing.**
2. **VS Code** - the editor where we write code.
3. **Git** (git-scm.com) - click Next with default options.

Check in the terminal. Both must show a version number:

```
python --version
git --version
```

3. One-Time Git Setup

Tell Git who you are. Do this only once:

```
git config --global user.name "Your Name"
git config --global user.email "your-email"
```

Use the same email as your GitHub account.

Write **user.name** and **user.email** exactly like this. Only the value inside quotes changes.

4. Git Workflow

1. **git clone <link>** - copy the repo to laptop (first time only).
2. Write or change code in VS Code and save the file (Ctrl+S).
3. **git add .** - gets all changed files ready for the commit.
4. **git commit -m "message"** - makes the save point.
5. **git push** - uploads the commit to GitHub.

Remember: run git pull before starting work, so you have the latest code.

Then open the GitHub website and check that your file is there.

5. Python Basics

- **print("text")** shows text on the screen.
- **List** = items kept in order. Example: [1, 2, 3]
- **Dictionary (dict)** = key and value pairs. Example: {"name": "Ravi"}
- **Function** = reusable block of code. We make it using **def** and a name.
- **for loop** = repeats code for every item in a list.

- **if** = runs code only when the condition is true.

Practice code (file day1_practice inside the backend folder):

```
students = [
    {"name": "Ravi", "cgpa": 8.4},
    {"name": "Asha", "cgpa": 7.2},
    {"name": "Imran", "cgpa": 9.1},
]

def attendance_percent(attended, total):
    return attended / total * 100

for s in students:
    if s["cgpa"] > 8:
        print(s["name"], "has CGPA above 8")

print(round(attendance_percent(41, 60), 1))
```

Output:

```
Ravi has CGPA above 8
Imran has CGPA above 8
68.3
```

My task: add one more student, then print students with CGPA below 7.

```
for s in students:
    if s["cgpa"] < 7:
        print(s["name"], "has CGPA below 7")
```

6. Revision Questions

Q1. Difference between git pull and git push?

Pull downloads code from GitHub to my laptop. **Push** uploads code from my laptop to GitHub.

Q2. Difference between list and dict?

A **list** keeps items in order and we find them by position. A **dict** keeps key-value pairs and we find them by key.

Q3. What does git add . do?

It marks all changed files in the folder as ready for the next commit. **It does not save yet. Commit saves.**

7. Common Errors and Fixes

- **python not recognized** - reinstall, tick "Add to PATH".
- **git not recognized** - close VS Code and open it again.
- **Permission denied on push** - accept the GitHub invite first.
- **Push rejected** - run **git pull** first, then **git push**
- **Link typed alone** - it goes after **git clone**, same line.

8. Day 1 Checklist

- ☐ python and git versions are showing
- ☐ Invite accepted and repo cloned
- ☐ My first Python file (hello) ran successfully
- ☐ Practice output is correct
- ☐ My changes pushed, file visible on GitHub
- ☐ Handwritten notes photo pushed to GitHub

Important: Commit often. Always pull before you push.